Cybernetic Security

Special Tools in Securing Networks

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Lecture Agenda

Honeypots
Sandboxing
Server virtualization

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1. Honeypots

- Honeypot is a “lure” for attackers:
  - Imitating a real IT asset:
    - Application Server (mostly),
    - Router or switch
    - Firewall,
    - Infrastructure server (e.g. DNS),
    - Client device (workstation, laptop, mobile, …).

Honeypot Purpose

- Gathering information about threats:
  - From the internet,
  - And from the private network,
    - Depending on the honeypot location.
- From honeypot, the data about threats (attacks) are more relevant for the network and more detailed:
  - In comparison to "generalized" information from databases of security product providers:
    - Talos by Cisco: https://www.talosintelligence.com
    - Open Worldwide Application Security Project (OWASP) https://owasp.org
    - …

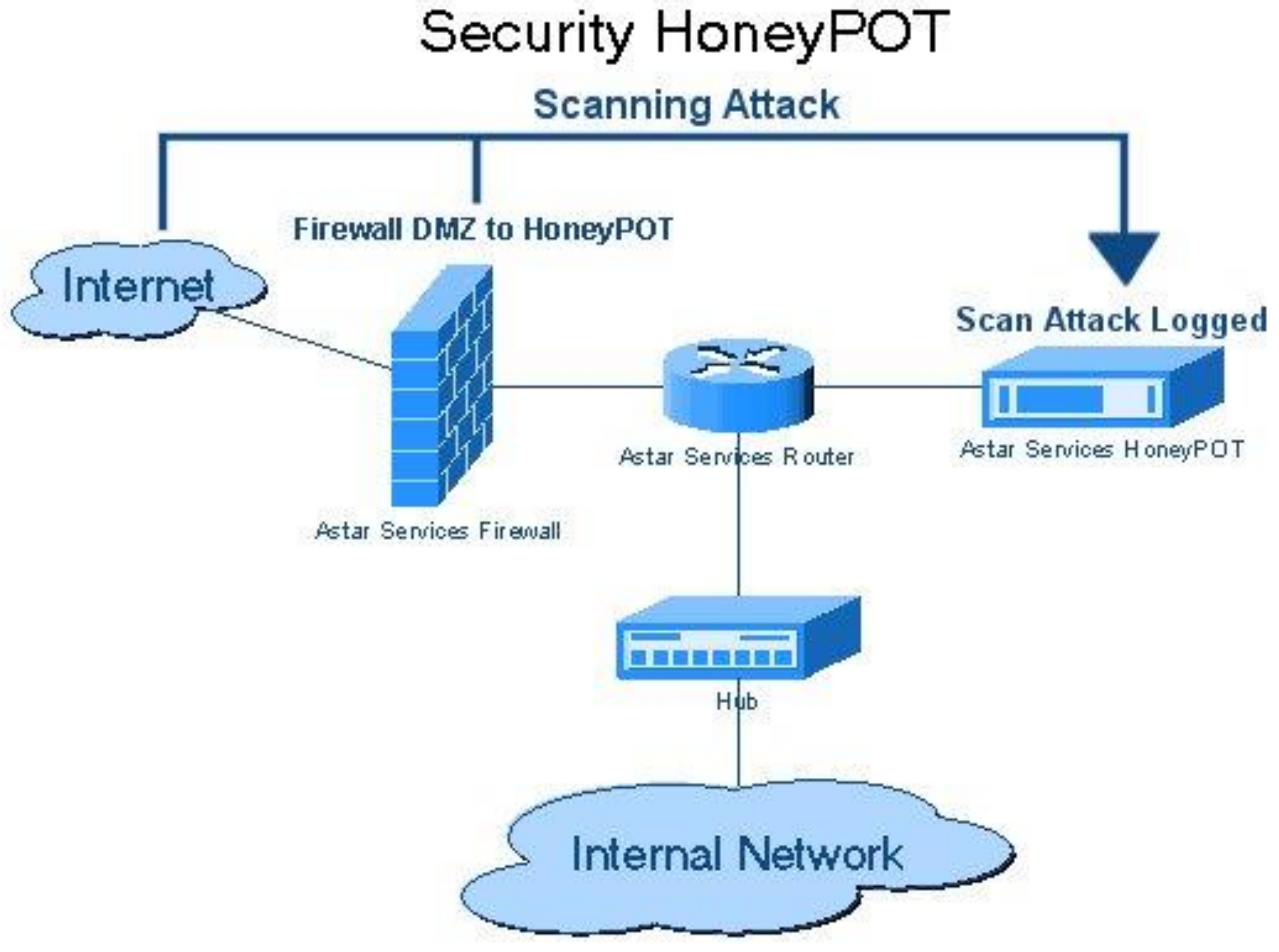
Honeypot Types

- According to the similarity to a real computer system:
  - Low-interaction honeypots:
    - Emulating just a single service (e.g. SSH for Linux servers).
  - Medium-interaction honeypots:
    - Emulation of multiple services,
    - Mode detailed emulation.
  - High-interaction honeypots:
    - Complex server is available for attackers (e.g. whole VM with full OS - limited outside)
- According to the honeypot activity:
  - Server honeypots (common, passive),
  - Client honeypots (actively searching for risky services, rare).

Complex Honeypot Structure: Components

- Specialized firewall (honeywall),
- Network device (router/switch) to decide:
  - What traffic will be passed to the network (production/private), and
  - What traffic shall go to the honeypot.
- Honeypot for capturing attacker's activities.
  - Capable to provide selected services:
    - SSH (Unix), SMB (Windows),
  - Capable to confine the attacker's activity inside the honeypot:
    - Network interaction with other networked nodes only for high-interaction honeypots.

Honeypot Diagram Example



Honeynets

- Multiple honeypots can cooperate.
  - Forming HONEYpot NETworks.
- Benefits of honeynets:
  - Sharing information about current attacks,
  - Better convincing attackers that interconnected servers are production server (not just lures for attackers).
    - Often used for high-interaction honeypots:
      - Multiple services/whole OS is available for attackers.

Honeypot Issues

- Attracting the traffic (including malicious),
- Deceiving attackers
  - That the honeypot is a real server worthy to be investigated/attacked.
  - e.g. capable of subsequent network interaction with other LAN components:
    - Like in case of malware spreading.
- Proper design of high-interaction honeypots.
- Analysis of captured attacks:
  - Time consuming activity,
  - Requires expertise in attack principles.

2. Sandboxing

Sandboxing is a technique for running a program/app in an isolated operating system (OS) environment so that:

- The software may not interact with other running software.
- The software may use only standard set of operating system API.
- The software is unable to inspect the OS.

- Sandboxing main purpose:
  - Provide an isolated execution environment to any application/software.

Sandboxing: Application

- Testing a malicious application behavior.
  - Antivirus: zero-day infections,
- Testing new applications:
  - From unknown developer, or
  - Before a new app is rolled out in a network (among users).

3. Server Virtualization

Virtual Machine (VM) is a SW container for running an operating system:

- with server applications/services/daemons.
- in the environment of a hypervisor.

- Different purpose from desktop virtualization:
  - Main goal: additional flexibility:
    - App running of physical server:
      - Any maintenance (e.g. OS or hardware update) required a server outage.
    - App running of virtual server:
      - Maintenance is performed on the underlying hardware.
      - Before a planned outage, running servers can be moved (migrated) to another physical server.

Server Virtualization Applications

- Omnipresent in on-premise servers:
  - Easier management,
  - Resource sharing: hardware cost reduction.
- Cloud services.
  - Application or service oriented.

Server Virtualization Issues

- Proper estimation of required (virtualized hardware) resources:
  - CPUs, RAM, disk space, …
  - Even in a software hypervisor, resources are assigned statically:
    - Only when a VM is not running, the resources can be reassigned (added, removed).
  - Server has to cope with traffic surges.
- Proper planning of outages:
  - For High Availability, 2 physical servers necessary:
    - migration to a backup (hot-standby) server before the outage.

Server Virtualization Alternatives

- Containerization:
  - E.g. Docker.
    - Lightweight containers run inside the same OS.
    - Platform as a Service (PaaS) approach.
    - Container runs shared OS components (services) once:
      - Unlike VMs.
    - Limited to a single OS and version.
- Server/cloud hosting:
  - HaaS/PaaS
    - Amazon cloud/Azure/Google cloud etc.

Conclusions

- Honeypot:
  - Protective measure to increase security.
- Sandboxing:
  - Protection of infrastructure.
- Server virtualization:
  - Saving resources for service provisioning.
  - Saving administration costs.

Questions?
Comments?

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Thank you for your attention

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